

First two steps in BSN method. (3)

Steps 1: Create Keys

f Compute a key for each record in the list by extracting relevant fields or portions of fields

f Effectiveness of the this method highly depends on a properly chosen key

Step 2: Sort Data

f Sort the records in the data list using the key of step 1

Any three nested-join names. (3)

1. Naive nested-loop join

2. Index nested-loop join

3. Temporary index nested-loop join

Any three parallel hardware architectures. (3)

f Symmetric Multi Processing (SMP)

f Distributed Memory or Massively Parallel Processing (MPP)

f Non-uniform Memory Access (NUMA)

A diagram of Processes A, B, C, D was given. Business impact on y-axis against Feasibility on x-axis. Which process would have the highest priority?

■ **Prioritization**

- Review & Prioritize findings
- Quadrant Technique

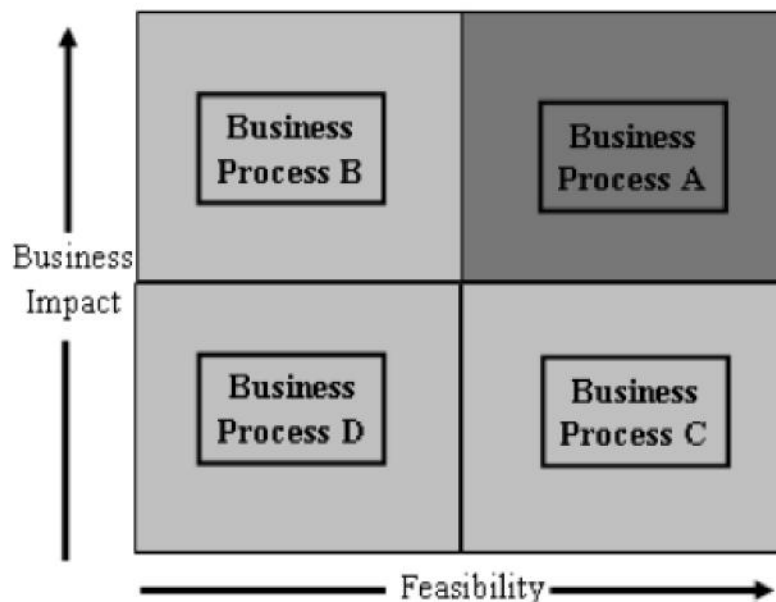


Figure -33.2: The quadrant method

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So the process having higher feasibility and impact is given higher priority over the process having lower feasibility and impact. In example of Figure 33.2, process A has highest priority while the process D has lowest priority.

In DWH Life Cycle, which is the most typical cycle used, even for the simplest of the processes? (3)

In the Data Warehousing (DWH) Life Cycle, the most typical cycle used, even for the simplest processes, is the Kimball Methodology.

In planning phase of DWH Lifecycle, define the key factors for the readiness of a project. (5)

- f Business sponsor
- f Business motivation
- f Feasibility
- f Business/IT relationship
- f Culture

List some reasons for why Web Warehousing may be important. (5)

1. Searching the web (web mining).
2. Analyzing web traffic.
3. Archiving the web.

The three reasons for warehousing web data are as listed in the slide.

Related to Quality Measurement Grid. A table was given. Write the name of the stage for the given situations. (5)

Quality Management Maturity Grid

	Management understanding	Quality organ. status	Problem handling	Cost of quality % of sales	Company attitude
Stage-1 Uncertainty	No comprehension of quality.	Quality Dept. part of manufacturing or engineering.	Fire fighting approach.	Reported unknown, actual high.	No organized activity.
Stage-2 Awakening	Recognize quality management may be of value.	Quality Dept. still part of manufacturing or engineering.	Short term solutions, no long term approach.	Reported as low, actually high.	All talk no real action.
Stage-3 Enlightenment	Become supportive and helpful.	Quality Dept. reports to top management.	Problems faced and solved orderly.	Reported as medium actually on the higher side.	Identifying and resolving problems
Stage-4 Wisdom	Understand absolutes of quality management.	Senior quality manager position.	Identified at an early stage.	Reported as about medium actually about medium.	Defect prevention is a routine.
Stage-5 Certainty	Quality management essential part of company policy.	Quality manager on board of directors.	Identified and resolved at an early stage.	Reported as low, actually is low.	Know why there are problems.

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Lexical error and Missing attributes ki definition lekhni thi.

Lexical errors:

For example, assume the data to be stored in table form with each row representing a tuple and each column an attribute. If we expect the table to have five columns because each tuple has five attributes but some or all of the rows contain only four columns then the actual structure of the data does not conform to the specified format.

Missing Attribute

- f Result of omissions while collecting the data.
- f A constraint violation if we have null values for attributes where NOT NULL constraint exists.
- f Case more complicated where no such constraint exists.
- f Have to decide whether the value exists in the real world and has to be deduced here or not.

Statistics and Mining ?

Data Mining Vs. Statistics

- f Formal statistical inference is assumption driven i.e. a hypothesis is formed and validated against the data.
- f Data mining is discovery driven i.e. patterns and hypothesis are automatically extracted from data.
- Said another way, data mining is knowledge driven, while statistics is human driven.

Three deliverables k names lekhni thi education Sy related tha question.

- f Data
- f Analytic applications
- f Data access tools
- f Education tools (1 to 2 days of development for each hour of education)

Five names of Development Methodology? Just name

Waterfall Model
Spiral Model
RAD
Structured Development
Data-Driven Methodologies
Goal-Driven Methodologies
User-Driven Methodologies

Limitation of Using Time contiguous log entries.?

- This method breaks down for visitors from large ISPs because different visitors may reuse dynamically assigned IP addresses over a brief time period.
- Different IP addresses may be used within the same session for the same visitor.
- This approach also presents problems when dealing with browsers that are behind some firewalls.

Shared Nothing RDBMS & partitioning k just Name?

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- f* Hash partitioning
- f* Key range partitioning.
- f* List partitioning.
- f* Round-Robin
- f* Combinations (Range-Hash & Range-List)

Cluster indexing

- f* A Cluster index defines the sort order on the base table.
- f* Ordering may be strictly enforced (guaranteed) or opportunistically maintained.
- f* At most one cluster index defined per table.
- f* Cluster index may include one or multiple columns.
- f* Reduced I/O.

Data Quality Validation Techniques

- f* Referential Integrity (RI).
- f* Attribute domain.
- f* Using Data Quality Rules.
- f* Data Histograming.

Classes of anomalies (lexical error and missing record)

3 Classes of Anomalies

Syntactically Dirty Data

- f* Lexical Errors
- f* Irregularities

Semantically Dirty Data

- f* Integrity Constraint Violation
- f* Business rule contradiction
- f* Duplication

Coverage Anomalies

- f* Missing Attributes
- f* Missing Records

What are two main types of Clustering

Two main types are one way clustering and two way clustering. One way clustering is when only data records (rows) are used. Two way clustering is when all the rows and columns are being used for clustering purpose.

Three dimensions in the context of web data Warehouse

Page, Event, Session, Referral

Three track names

- 1 Lifecycle Technology Track
- 2 Lifecycle Data Track
- 3 Lifecycle Analytic Applications Track

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Development methodologies

- Waterfall model
- Spiral model
- RAD Model
- Structured Methodology
- Data Driven
- Goal Driven
- User Driven

Cluster index ka el table diya tha osko cluster mei krna th

Cluster Index: Example

Student	Name	Age	Campus	Tech
s9	afridi	19	Lahore	CS
s1	amir	20	Lahore	Elect
s2	javed	20	Islamabad	CS
s4	imran	20	Peshawar	Elect
s5	majid	20	Karachi	Telecom
s3	salim	21	Lahore	CS
s7	tahir	21	Peshawar	Telecom
s6	taslim	25	Karachi	CS
s8	sohaib	26	Peshawar	CS

Cluster indexing on AGE

One indexing column at a time

Student	Name	Age	Campus	Tech
s9	afridi	19	Lahore	CS
s2	javed	20	Islamabad	CS
s3	salim	21	Lahore	CS
s6	taslim	25	Karachi	CS
s8	sohaib	26	Peshawar	CS
s1	amir	20	Lahore	Elect
s4	imran	20	Peshawar	Elect
s5	majid	20	Karachi	Telecom
s7	tahir	21	Peshawar	Telecom

Cluster indexing on TECH

Data quality analysis project

For each data quality analysis project selected, the data quality project manager defines the scope of the project and defines the level of analysis that will be the most beneficial for the project under question.

Find cost of hash in matching phase.

Cost of Hash-Join

- In partitioning phase, read + write both operations requires $2(M+N)$ I/Os.
- In matching phase, read both requires $M+N$ I/Os.

DWH lifecycle phases

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DWH Lifecycle- Step 1: Project Planning
DWH Lifecycle- Step: 2 Requirements Definition
DWH Lifecycle- Step 3.1: Technology Track
DWH Lifecycle- Step 3.3: Analytic Applications Track
DW Lifecycle- Step 4: Deployment
DW Lifecycle- Step 4: Deployment

Sources of web scraping also explain three reasons

Web scrapping

Lot of data in a web page, but is mixed with a lot of “junk”.

Problems:

- f Limited query interfaces
- f “Free text” fields
- f Inconsistent output
- f Rapid change without notice.

Dens and spiral indexing

Dense Index: Every key in the data file is represented in the index file

Sparse Index: A sparse index uses less space at the expense of somewhat more time to find a record given its key.

11110001100 Is trha ka tha is ki output btani thi

Case-1

1111000011110000001111100000011111 INPUT
14#04#14#06#15#06#15 OUTPUT

Case-2

1010101010101010101010101010101010 INPUT
11#01#11#01#11#01#11#01#... OUTPUT

Case-3

11111111111111100000000000000000 INPUT
117#017 OUTPUT

Data profiling

Data profiling is a process which involves gathering of information about column through execution of certain queries with intention to identify erroneous records.

DTS m koi items use Hoti hn UN k name puchy thy

DTS Tasks

- **DTS Package contains one or more tasks**
- **Task defines single work item**
 - Establishing connections
 - Importing and exporting data
 - Transforming data
 - Copying database objects
 - etc

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Kimball lifecycle 5 phases k name

DWH Lifecycle: Key steps

1. Project Planning
2. Business Requirements Definition
3. Parallel Tracks
4. Deployment
5. Maintenance

Supervised learning or unsupervised learning techniques ko side py krna tha

Supervised learning is when you are performing DM the supporting information that helps in the DM process is also available. When we will talk about unsupervised learning you will not have such known or a priori knowledge. In other words you can not give such factors as input to the DM technique which can facilitate your DM process. So wherever the user gives some input that is supervised else that is unsupervised learning.

Kimball diagram thi us k kuch statements missing thin Wo btain thin 5numbers ka

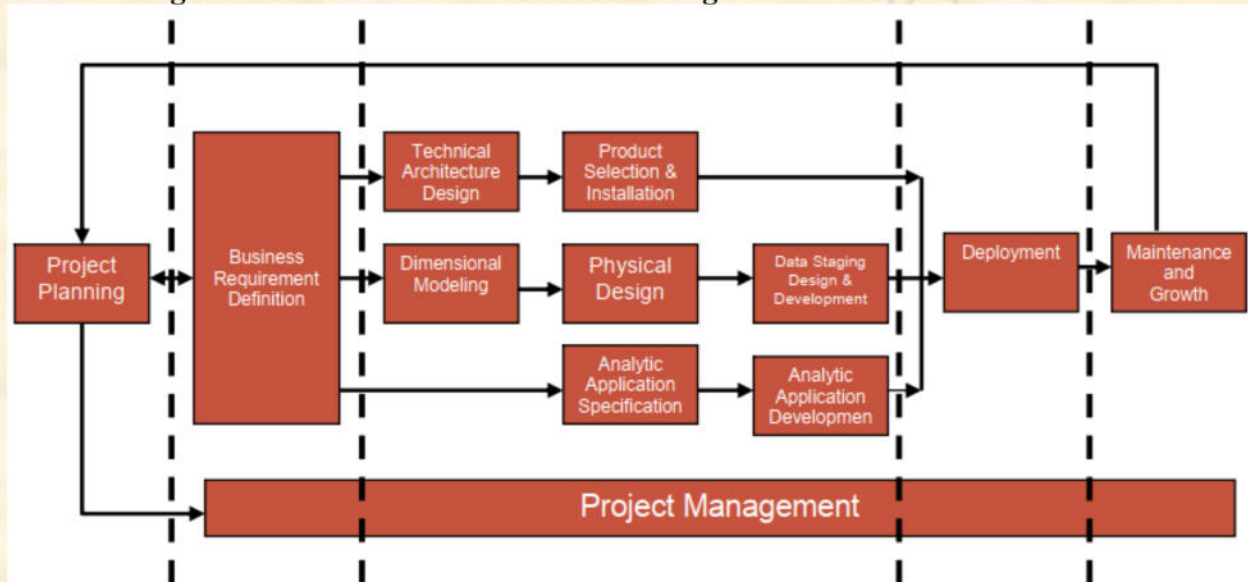


Figure -33.1: Business Dimensional Lifecycle (Kimball's Approach)

Web data kahan sy liya jata hai

Web data is collected from various sources on the internet, including websites, social media platforms, online databases, and search engines. Web scraping is a common technique used to extract data from websites

Aik program tha header ka btana th

Data from Karachi Campus: Sample



	A	B	C	D	E
1	St_ID	Name	Father	DoB	Gender
2	0	Mahjabeen Paracha	Salim Abdul kareem Paracha	16-Jun-79	F
3	1	Abdul Wasay Awan	Ch. Abdur Rahman Awan	13-Dec-77	M
4	2	Kina Zafar	Zafar Kamal	13-Sep-79	F
5	3	Sultan Jabir Sherazi	Sohail Sherazi	15-Sep-79	M

Karachi: Header of Student Table

Web page dimension. list any 5 attributes of this dimension.(5)

- Page key
- Page source
- Page function
- Page template
- Item type

Pk or non pk btana tha

The problems associated with the data can correspond to either the primary keys or non primary keys, as the nature of the problem and the corresponding solution varies with the type of the key involved.

Hash based ya b tree kon sa kab use hota or SQL query likhni the us ke

B-tree vs. Hash Indexes

f Indexing (using B-trees) good for range searches, e.g.:

SELECT * FROM R WHERE A > 5

f Hashing good for match based searches, e.g.:

SELECT * FROM R WHERE A = 5

Parallel Software Architectures

- f Shared Memory
- f Shard Disk
- f Shared Nothing

Nested-Loop Join: Cost Formula

Join cost = Blocks accessed for Table_A +

Blocks accessed for Table_A × Blocks accessed for Table_B

How to improve Data Quality?

The four categories of Data Quality Improvement

- f Process
- f System

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f Policy & Procedure
f Data Design

A simplified ERD of the Agri System

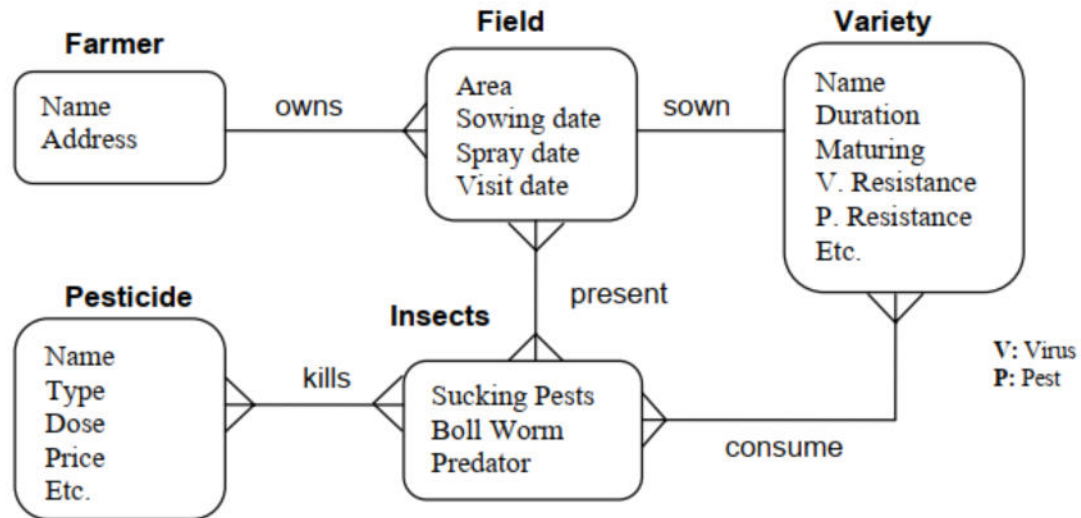


Figure-37.5: A simplified ERD of the Agri System being considered